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Professional Summary

Applied Scientist at Amazon specializing in context-aware **LLM mid/post-training and rubric-based evaluation** for **agentic systems** — including context extension, grounding, personalization, and multi-turn consistency — on **100B+ MoE** backbones, with experience productionizing **closed-loop post-training data infrastructure** that turns live-traffic failures into continuous model improvements. **Ph.D. in Computer Science** with **20+ publications** (8 first-authored) at **NeurIPS (Spotlight)**, ACL, EMNLP, and NAACL, including **early contributions** to **multi-agent systems and set-attention** for tabular language models, one **US patent**, and area chair at **ARR**.

Experience

Amazon 2024.1 – Present
Applied Scientist (full-time) Bay Area, CA

- **Rufus Foundation Modeling (startup-like) — Context-aware Mid/Post-training & Evaluation:** Drove the context management and RAG workstream for shopping agents on 100B+ MoE backbones.
 - Extended long-context capability from **8K to 128K** tokens through **continual pre-training (ABF, YaRN)**, with training-free methods supporting up to **1M** tokens, enabling stable context window over extended multi-turn sessions.
 - Built a **verifiable rubric-based eval** for personalization helpfulness, raising human alignment from **40% → 78%** over 5 iterations; used the eval to drive **GRPO** optimization, lifting helpfulness from **84% → 93%** under user-specific preference settings.
 - Increased factual accuracy from **67% → 90%** on customer-facing RAG evaluations by building **synthetic IFT + DPO pipelines** for faithfulness and grounding.
 - Enhanced multi-turn context consistency by incorporating live-traffic failure patterns into **DPO** training, improving context carryover from **61% → 73%**.
- **Customer Service — Post-training Data Infrastructure & Flywheel:** Owned the **end-to-end post-training data flywheel** for the team's multi-agent LLM system.
 - Built a **live-traffic failure mining and analytics system** that surfaced actionable patterns directly feeding post-training data generation (e.g., tone-related issues driving 5% of weekly failures).
 - Productionized a scalable **annotation and quality control** system on Amazon Bedrock for **SFT/DPO/GRPO /retrieval/eval** data, with sample-level validation (rule-based + LLM-as-judge), dataset-level controls (diversity, de-duplication), and LakeFS-based versioning — adopted across teams as the standard data pipeline.
 - Developed **diversity-aware data selection**: agent tool-use accuracy **86% → 93%**, training time **4× faster (24h → 6h)**.
 - Introduced **difficulty-aware** curriculum data selection: response generation quality **88% → 91%**, training time **5× faster (10d → 2d)**.
- **Intern Mentorship:** Mentored two interns on LLM research projects in long-context modeling and multi-turn improvement with verifiable GRPO — **2 papers published, 1 under review**.

Amazon Web Services (AWS) 2023.5 – 2023.8
Applied Scientist (Intern) Santa Clara, CA

- Developed a **collaborative multi-agent** prompting framework (**CoMM**) achieving SOTA on complex science benchmarks in zero-shot and few-shot settings — published at NAACL 2024.

Amazon Web Services (AWS) 2022.6 – 2023.1
Applied Scientist (Intern) Santa Clara, CA & remote

- Developed **HYTREL**, a **hypergraph-based tabular LM** capturing structural dependencies, achieving SOTA on tabular representation learning — published as **NeurIPS 2023 Spotlight (top 5%)**.

Tencent AI Lab
NLP Researcher (Intern)

2021.6 – 2021.8
remote (College Station, TX)

- Built a benchmark for zero-shot knowledge base completion across diverse KB sources — published at ICDM 2022 KG Workshop.

National Lab of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

2018.1 – 2019.7

Research Engineer (Intern & Full-time)

Beijing, China

- Built information extraction and evaluation systems for financial event extraction and causality detection.

Selected Publications

LLM & Foundation Model Training — *Data-centric and structure-aware approaches to LLM training and alignment.*

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. “HYTREL: Hypergraph-enhanced Tabular Data Representation Learning.” *NeurIPS 2023, Spotlight (5%), first work to use set-attention for tabular LMs.*
- Ming Li, **Pei Chen**, Chenguang Wang, et al. “Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning.” *ACL 2025, Findings.*
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. “END: Early Noise Dropping for Efficient and Effective Context Denoising.” *pre-print.*
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. “ItD: Large Language Models Can Teach Themselves Induction through Deduction.” *ACL-2024.*
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. “Aligning Large Language Models with Implicit Preferences from User-Generated Content.” *ACL 2025.*

Agent — *Multi-agent reasoning, agent-oriented pre-training, and verifiable tool-use training.*

- **Pei Chen**, Shuai Zhang, Boran Han. “CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving.” *NAACL 2024, early work in demonstrating multi-agent systems for complex LLM reasoning.*
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. “Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training.” *NAACL 2025.*
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. “Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates.” *EMNLP 2025.*
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, **Pei Chen**, et al. “Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intents.” *accepted to ACL 2026.*

RAG & Multi-turn & Personalization — *Context-aware conversational LLMs across long-context, retrieval, and personalization modeling.*

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. “LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios.” *NAACL 2025.*
- Ming Li, **Pei Chen**, Zhenhao Zhang, et al. “Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.” *Accepted to ACL 2026.*
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. “ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations.” *NAACL 2025.*
- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. “UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations.” *ACL 2025.*
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. “Leveraging historical information to boost retrieval-augmented generation in conversations.” *Information Processing & Management, March 2026.*
- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. “Instant Personalized Large Language Model Adaptation via Hypernetwork.”, *accepted to ACL 2026.*
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. “POPI: Personalizing LLMs via Optimized Natural Language Preference Inference.” *pre-print.*

Information Extraction — *Graph-based modeling of non-sequential dependencies in structured extraction (Ph.D. work).*

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition.” ACL-IJCNLP 2021.
- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. “Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition.” NAACL 2022.
- **Pei Chen**, Hang Yang, Kang Liu, et al. “Reconstructing Event Regions for Event Extraction via Graph Attention Networks.” AACL-IJCNLP 2020.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Domain-specific Named Entity Recognition via Graph Neural Networks.” US Patent 12,530,529, 2026.

Education

Texas A&M University <i>Ph.D. in Computer Science</i>	2019.8 - 2024.5 College Station, TX
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- **GPA:** 4.0/4.0
- **Research Directions:** Information extraction and structured NLP (NER, event extraction, knowledge base completion); later extended to LLMs and agentic systems.
- **Dissertation:** Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions.
- **Teaching assistant:** CSCE 636 Deep Learning and CSCE 313 Computer Systems.

Southwestern University of Finance and Economics <i>Master's Degree in Finance</i>	2016.9 – 2018.6 Chengdu, China
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- **GPA:** 3.9/5.0, ranking 1/178.
- **Awards:** 2017 National Scholarship for Graduate Student.

National University of Defense Technology <i>B.Eng. in Simulation Engineering</i>	2010.9 – 2014.6 Changsha, China
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- **GPA:** 88.61/100, ranking 1/45.
- **Thesis:** Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.

Professional Service

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- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
 - 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
 - 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
 - 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
 - 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
 - 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

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- **Training Frameworks:** Megatron, Verl, Slime
 - **Post-training Methods:** Continual Pre-training/Mid-training (ABF, YaRN), SFT, DPO, GRPO, GSPO, DAPO
 - **Evaluation Methodology:** Verifiable rubric design, human alignment calibration (iterative refinement), LLM-as-judge validation, synthetic eval data generation
 - **Programming:** Python, PyTorch, SQL, C/C++, Git, Docker